Name: ____

Forces and motion 1

higher

Date:

Time:

Total marks available: 59

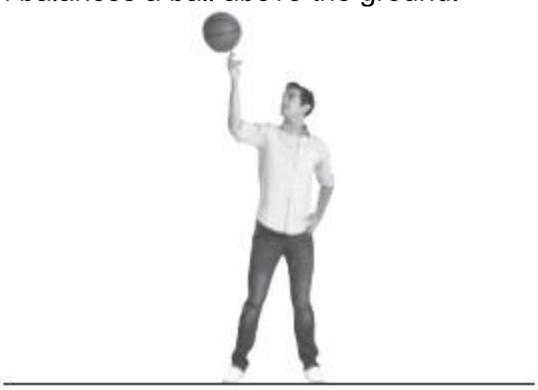
Total marks achieved: ______

KSA

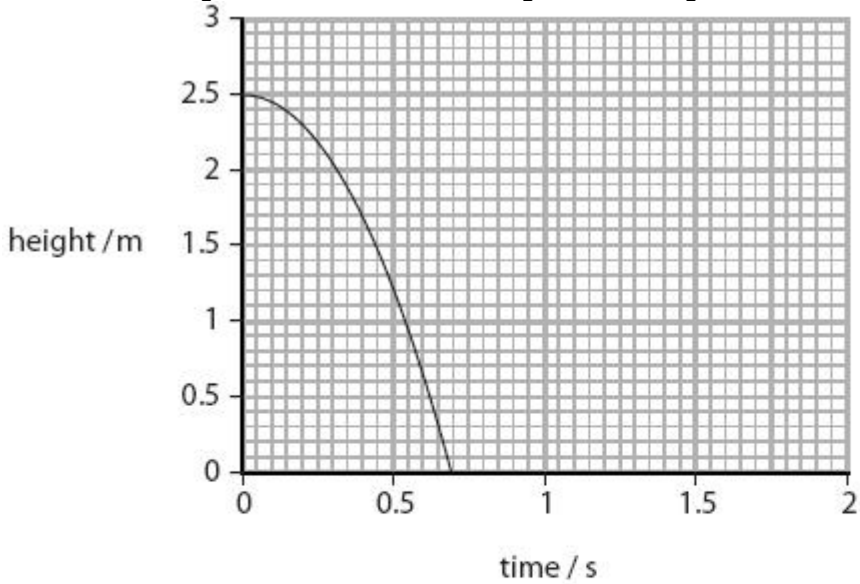
Questions

Q1.

The man in the photograph balances a ball above the ground.



He lets the ball fall.
He starts a timer at the same time.
The graph shows how the height of the ball above the ground changes with time.



(i) From the graph, state the height of the ball above the ground when the timer was started.

(1)

....

(ii) From the graph, state the time taken for the ball to reach the ground.

(1)

....

(iii) The ball bounces back to a height of 1.9 m.
Continue the line on the graph to show this.

(3)

(iv) Explain why the ball does not bounce back to its original height.

(2)

....

....

....

....

Q2.

Figure 4 shows two students investigating their reaction times.

Student B supports his left hand on a desk.

Student A holds a ruler so that the bottom end of the ruler is between the finger and thumb of student B.

When student A releases the ruler, student B catches the ruler as quickly as he can.

The investigation is repeated with the right hand of student B.

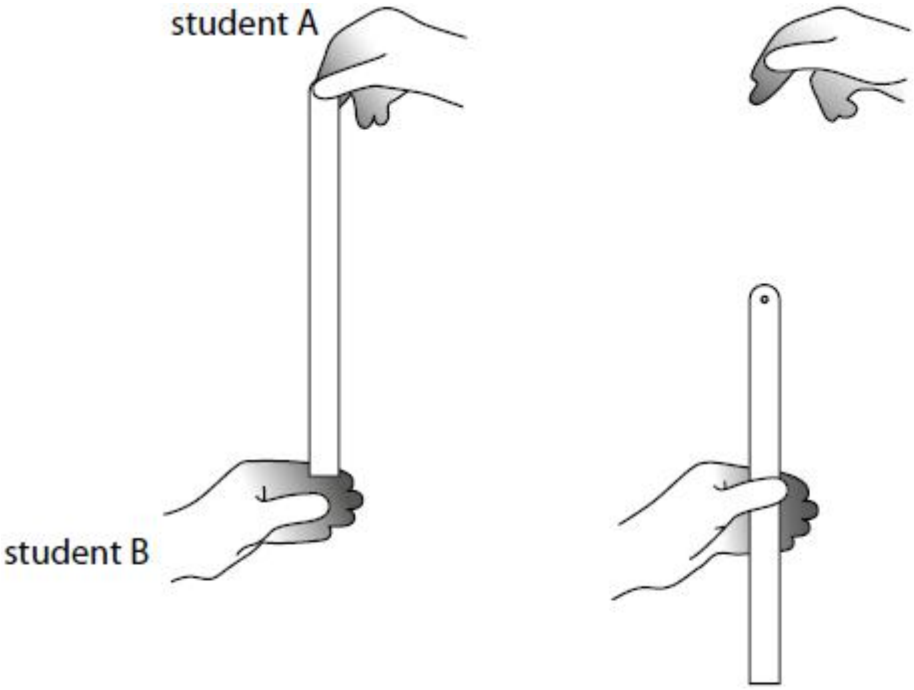


Figure 4

(a) The students took five results for the left hand and five results for the right hand.

Figure 5 shows their results.

which hand	distance dropped (cm)					
	trial 1	trial 2	trial 3	trial 4	trial 5	average
left	10.1	25.5	18.4	14.6	11.7	14
right	17.5	16.1	19.4	18.6	20.2	....

Figure 5

(i) Calculate the average distance dropped for the right hand.
Give your answer correct to 2 significant figures.

(2)

distance = ... cm

(ii) Calculate the average time for the left hand.
Use the equation

$$\text{time}^2 = \frac{\text{distance}}{500}$$

(2)

average time = ... s

(b) Explain whether any of the readings are anomalous.

(2)

....

....

....

....

(c) Give **two** ways that the students can improve the quality of their data other than ignoring anomalous results.

(2)

1 ....

....

2 ....

....

(d) Describe how the students could develop their investigation to investigate how reaction time changes with another variable.

(2)

....

....

....

....

....

....

(Total for question = 10 marks)

Q3.

Figure 10 shows a footballer and a ball.

(Source: © GelgelNasution/Shutterstock)

Figure 10

The ball is stationary before it is kicked.

The mass of the ball is 0.43 kg.

The kick exerts a force of 2700 N on the ball.

The velocity of the ball just after it has been kicked is 36 m/s.

(i) Calculate the momentum of the ball just after it has been kicked.

State the unit for momentum.

(3)

momentum = ...

unit ....

(ii) Calculate the amount of time that the footballer's kicking foot is in contact with the ball.

(2)

time = ... s

(iii) Explain how Newton's third law applies to the interaction between the kicking foot and the football.

(3)

....

....

....

....

....

....

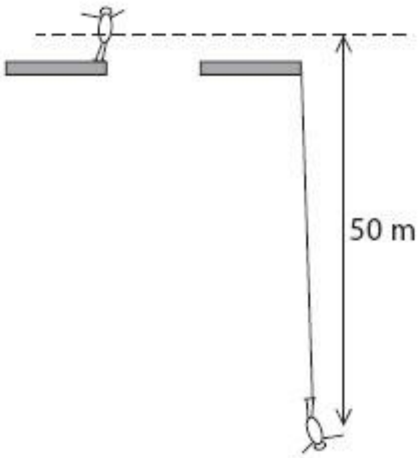
....

....

(Total for question = 8 marks)

Q4.

A 60 kg student weighs 600 N.
He does a bungee jump.



The bungee cord becomes straight and starts to stretch when he has fallen 50 m.
Complete the sentence by putting a cross (☒) in the box next to your answer.

When his speed is 10 m/s his momentum is

(1)

☐ **A** 600 kg m/s

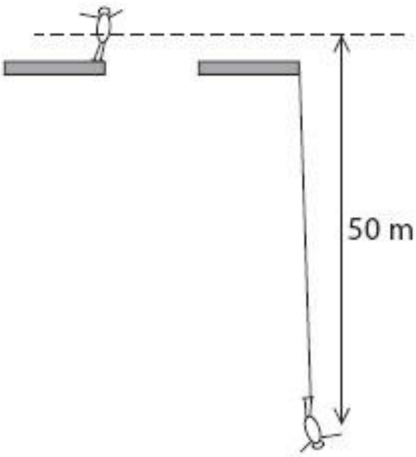
☒ **B** 3 000 kg m/s

☒ **C** 6 000 N m/s

☒ **D** 30 000 N m/s

Q5.

A 60 kg student weighs 600 N.
He does a bungee jump.



The bungee cord becomes straight and starts to stretch when he has fallen 50 m.
(a) Complete the sentence by putting a cross (■) in the box next to your answer.

He first stops moving

(1)

☒ **A** before all the energy has disappeared

☒ **B** before the bungee cord starts to stretch

☒ **C** when the bungee cord is stretched the most

☒ **D** when the elastic potential energy is zero

(b) Complete the sentence by putting a cross (■) in the box next to your answer.

When his speed is 10 m/s his momentum is

(1)

☒ **A** 600 kg m/s

☒ **B** 3 000 kg m/s

☒ **C** 6 000 N m/s

☒ **D** 30 000 N m/s

(c) (i) Calculate the change in gravitational potential energy as the student falls 50 m.
Give the unit.

(3)

....

(ii) State at what point in the bungee jump the student has maximum kinetic energy.

(1)

....

....

(iii) Explain why his maximum kinetic energy is likely to be less than your answer to (c)(i).

(2)

....

....

....

(Total for Question = 8 marks)

Q6.

Figure 6 shows a different velocity/time graph.

This straight line graph can be represented by the equation

$$y = mx + c$$

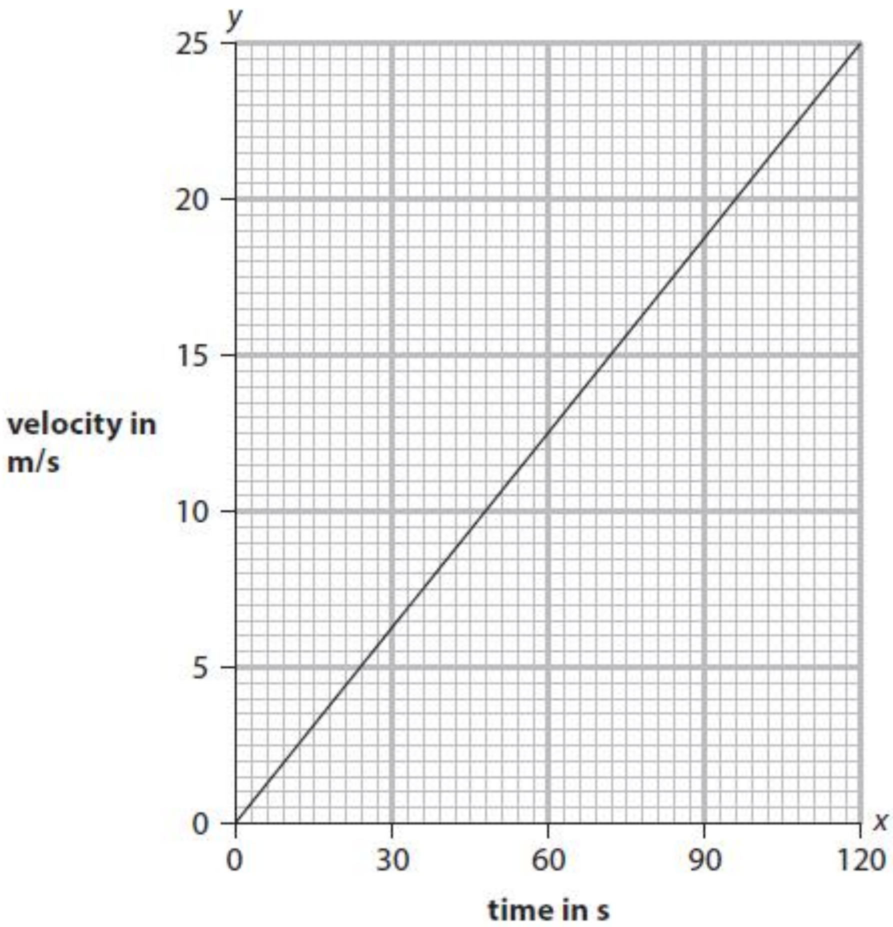


Figure 6

(i) Give the quantities that *x* and *y* represent in the equation.

(1)

x represents ....

y represents ....

(ii) Calculate the value of *m* from the graph in Figure 6.

(2)

m = ... m/s^2

(iii) State the value of *c* from the graph in Figure 6.

(1)

value of *c* = ...

(Total for question = 4 marks)

Q7.

A student has some cupcake cases.

One cupcake case is shown in Figure 1.

(Source: © Anton Starikov/Shutterstock)

Figure 1

The student drops a stack of cupcake cases with the base facing downwards, as shown in Figure 2.

(Source: © Elena Schweitzer/Shutterstock)

Figure 2

The speed of the falling stack of cupcake cases depends on the number of cupcake cases in the stack.

(i) The student also has a stop clock and a metre rule.

Describe an investigation to show how the speed of the falling stack of cupcake cases depends on the number of cupcake cases in the stack.

(4)

....

....

....

....

....

....

....

....

(ii) A stack of cupcake cases has a mass of 0.005 kg.

Calculate the weight, in newtons, of the stack of cupcake cases.
Gravitational field strength = 10 N / kg

(2)

Use the equation

$$W = mg$$

weight = ... N

Figure 3 shows a cupcake case that is falling at a constant velocity.

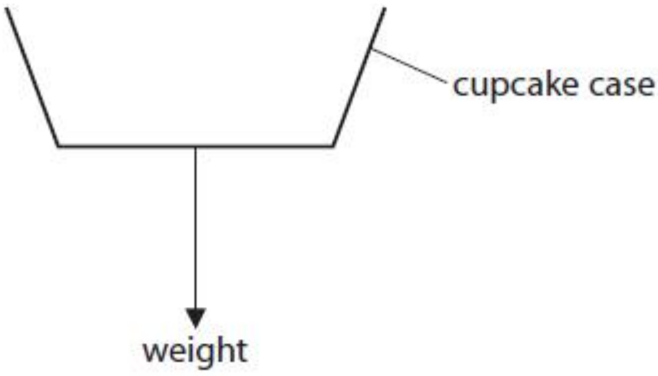


Figure 3

(iii) Draw an arrow on Figure 3 to show the force due to air resistance on the cupcake case.

(1)

(iv) State the value of the acceleration of the cupcake case when it is falling at a constant velocity.

(1)

....

(Total for question = 8 marks)

Q8.

Figure 10a shows a box falling towards a hard floor.

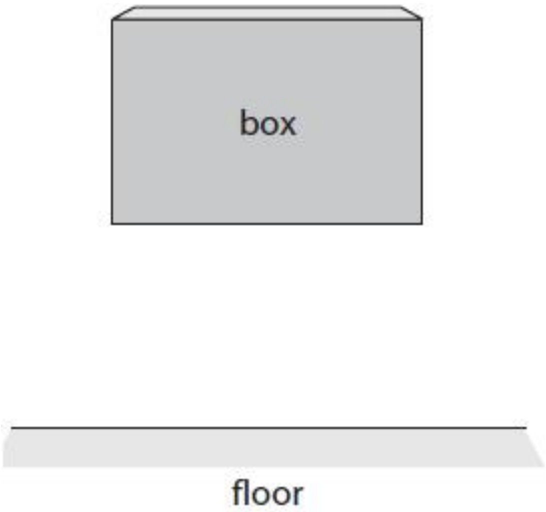


Figure 10a

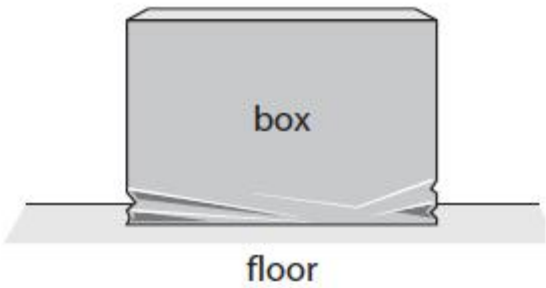


Figure 10b

The box hits the floor and crumples a little before it comes to rest as shown in Figure 10b.

The momentum of the box just before it hits the floor is 8.7 kg m/s.

The box comes to rest 0.35 s after it first hits the floor.

(i) Calculate the magnitude of the force exerted by the floor on the box.

Use an equation selected from the list of equations at the end of this paper.

(2)

force exerted by the floor on the box = ... N

(ii) State the magnitude and direction of the force exerted by the box on the floor.

(2)

magnitude ....

direction ....

(Total for question = 4 marks)

Q9.

Figure 8 shows two ice skaters during a performance.

Figure 8

(i) The two ice skaters are travelling together in a straight line at 3.50 m/s.

Their total momentum is 371 kgm/s.
The man has a mass of 64.5 kg.
Calculate the mass of the woman.

(4)

mass = ... kg

(ii) Calculate the kinetic energy of the man.

(2)

kinetic energy = ... J

(Total for question = 6 marks)

Q10.

Figure 5 is a velocity/time graph for a lift moving upwards in a tall building.

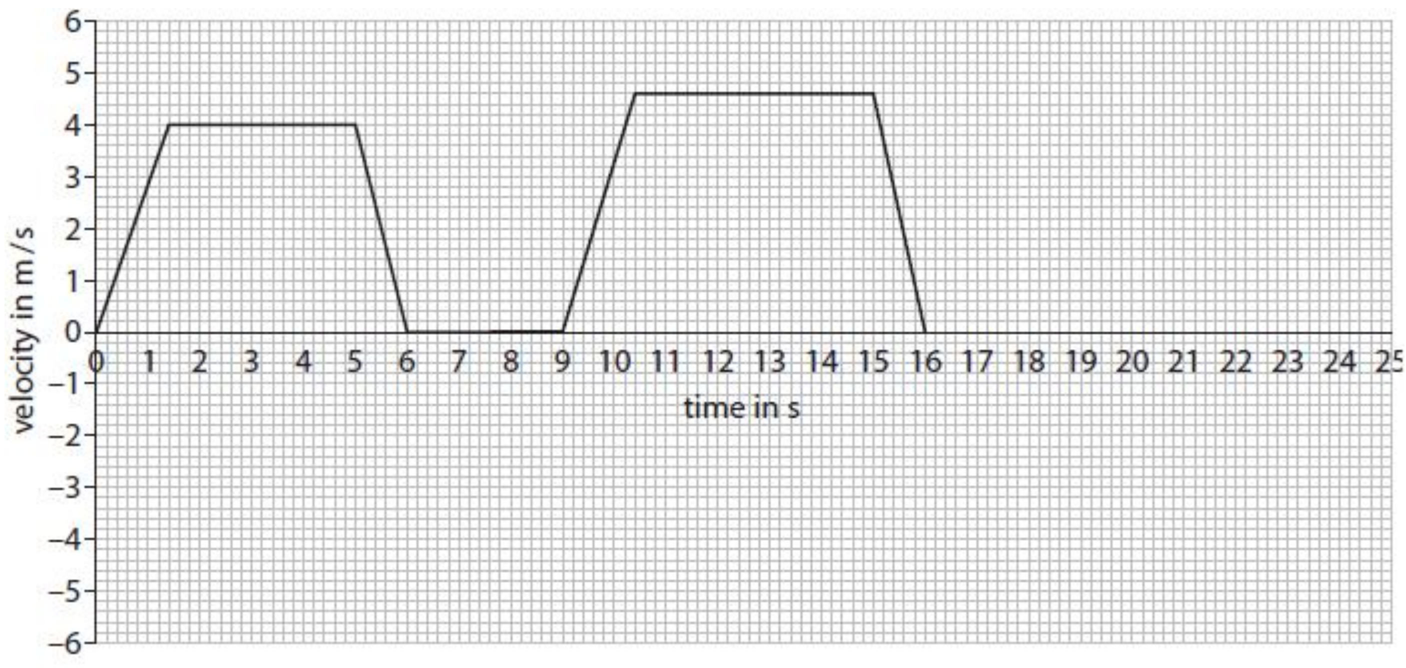


Figure 5

Use the graph in Figure 5 to determine the acceleration of the lift during the first 1.4 s.

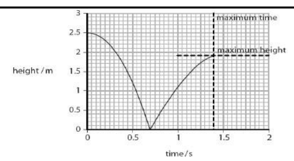
(3)

acceleration = ... m/s^2

(Total for question = 3 marks)

Mark Scheme

Q1.

	Answer	**Acceptable answers**	**Mark**
(i)	2.5 (m)	Allow answers between (and including) 2.45 & 2.55	**(1)**
(ii)	0.7 (s)	Allow answers between (and including) 0.68 & 0.72	**(1)**
(iii)	 line: same shape as	Ignore any part of the graph after the peak	**(3)**

	original (1) peak at 1.9 m (1) time taken < 0.7 s (1)		
(iv)	An explanation linking: energy lost (1) in collision with ground / air resistance (1)	Inelastic collision worth (2) as sound or heat	**(2)**

Q2.

Question number	Answer	Additional guidance	Mark
(a)(i)	Calculating the mean (1) 18.36 Rounding to 2 s.f. (1) 18 (cm)	award full marks for correct numerical answer without working	(2)

Question number	Answer	Additional guidance	Mark
(a)(ii)	Rearrangement (1) $t = \sqrt{\frac{\text{distance}}{500}}$ Substitution and answer (1) time = 0.17 (s)	award full marks for correct numerical answer without working allow answers which round to 0.17, e.g. 0.1673	(2)

Question number	Answer	Additional guidance	Mark
(b)	An explanation that combines identification via a judgement (1 mark) to reach a conclusion via justification/reasoning (1 mark): • 25.5 is an anomalous result (1) • (because) it is much further away from the mean than the other results (1)	ignore 19	(2)

Question number	Answer	Mark
(c)	• Take more readings (1) • Idea that a third student should also measure the reaction time (1)	(2)

Question number	Answer	Additional guidance	Mark
(d)	An answer that combines the following points to provide a logical description of the plan/method/experiment: • using a larger group of students/large population of students (1) • and measure how their reaction time varies with age/height (1)	allow any suitable variable	(2)

Q3.

Question number	Answer	Additional guidance	Mark
(i)	selection of (p =) m × v (1) evaluation (1) (momentum =) 15 unit (1) kg m/s	(momentum =) 0.43 × 36 allow numbers that round to 15 e.g. 15.48 allow 15.5 award 2 marks for the correct value without working kg m s^{-1} allow N s	(3) AO2.1

Question number	Answer	Additional guidance	Mark
(ii)	selection of force = $\frac{\text{change in momentum}}{\text{time}}$ (1) evaluation (1) 0.0056 (s) ALTERNATIVE METHOD selection of F=ma **and** a=(v-u)/t (1) evaluation (1) 0.0057 (s)	$2700 = \frac{15}{\text{time}}$ OR $2700 = \frac{\text{their (a)(i)}}{\text{time}}$ OR $(\text{time}) = \frac{15}{2700}$ OR $(\text{time}) = \frac{\text{their (a)(i)}}{2700}$ allow numbers that round to 0.0056 e.g. 0.005555 (using Δp=15) or numbers that round to 0.0057 e.g. 0.005733 (using Δp=15.48) allow ECF from their (a)(i) for 2 marks without working allow numbers that round to 0.0057 e.g. 0.00573 award full marks for the correct value without working	**(2)** **exp** **AO2.1**

Question number	Answer	Additional guidance	Mark
(iii)	explanation linking equal **and** opposite forces (1) (which are) force exerted by the foot (on the ball) (1) (and) force exerted by the ball (on the foot) (1)	accept action and reaction are equal and opposite (which are) force exerted (by the foot) on the ball (1) (and) force exerted (by the ball) on the foot (1) award 2 marks for force on ball = force on foot award 3 marks for force on ball = force on foot but in opposite direction ignore references to other named forces and energy	(3) AO2.1

Q4.

	Answer	Acceptable answers	Mark
	A 600 kg m/s		**(1)**

Q5.

	Answer	Acceptable answers	Mark
(a)	**C** when the bungee cord is stretched the most		**(1)**
(b)	**A** 600 kg m/s		**(1)**
(c)(i)	Substitution: (1) 60 × 10 × 50 or 600 × 50 Evaluation: (1) 30 000 Unit: (1) J / Nm	give two marks for correct answer no working j / joule 30 kJ for full marks	**(3)**
(c)(ii)	After falling 50 m / when the cord becomes	tension starting to increase at terminal velocity	**(1)**

	straight/when cord starts to stretch	ignore maximum velocity/speed	
(c)(iii)	An explanation linking any two of not all GPE is transferred to KE (1) some of the GPE transfers to thermal energy /work is done (1) due to drag (1)	not all GPE goes to KE maximum energy is same (value) as GPE before falling /speed does not reach the speed at which he should fall some lost as heat/sound (of rope or movement through air) (air) resistance / friction ignore wind	**(2)**

Q6.

Question	Answer	Additional guidance	Mark
(i)	x represents **time** OR **t** y represents **velocity** OR **v**	both correct in correct place for 1 mark ignore units	**1 AO1.1**

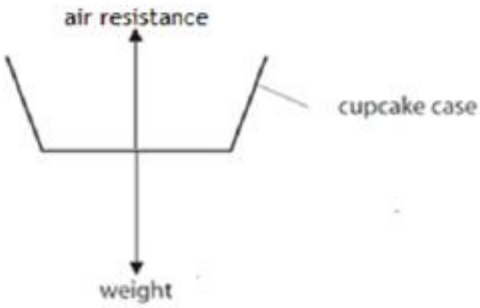
Question	Answer	Additional guidance	Mark
(ii)	attempt to find a gradient of the line that would give an answer between 0.18 and 0.24 (1) evaluation(1) 0.21 (m/s^2)	e.g. $\frac{12.5}{60}$ or $\frac{10}{42}$ values that are between 0.20 and 0.22 (m/s^2) e.g. 0.2083 do not allow fractions in the answer line for evaluation mark award full marks for the correct answer without working	**2** **AO2.1**

Question	Answer	Additional guidance	Mark
(iii)	0	zero	**1** **AO2.1**

Q7.

Question number	Answer	Additional guidance	Mark
(i)	A description to include any 4 from: measure height (1) measure time of fall (1) use (average) speed=distance ÷ time (1) repeat with different number of cupcake cases in the stack/more cupcake cases (1) repeat **and** average time (of fall for each stack of cupcake cases) (1) plot a graph (speed of fall against number of cupcake cases dropped) (1)	 allow 'keep same height' allow in this context hold against (fixed point on) metre rule allow 'time it' accept cupcakes for cupcake cases	**(4)** **AO1**

Question Number	Answer	Additional guidance	Mark
(ii)	substitution (1) (W=)0.005 x 10 evaluation (1) 0.05 (N)	 5×10^{-2} (N) do not allow power of ten error award full marks for the correct answer with no working give full credit for use of g=9.8 or 9.81 N/kg	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(iii)	 air resistance arrow (1)	Judge by eye any vertical upward arrow outside or inside the cupcake case ignore length of arrow arrow need not touch cupcake holder ignore label on arrow	**(1)** **AO2**

Question number	Answer	Additional guidance	Mark
(iv)	zero / there is none / 0 / it has no acceleration	ignore 'constant' ignore units	**(1)** **AO2**

Q8.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) (force =) $\frac{8.7}{0.35}$ evaluation (1) 25 (N)	use of force = $\frac{\text{change in momentum}}{\text{time}}$ allow numbers that round to 25 e.g 24 .8571 award full marks for correct answer without working.	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(ii)	(magnitude) 25 (N) (1) (direction) down(wards)/ towards floor (1)	ecf from i allow arrow drawn pointing down "south"	(2) AO3

Q9.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) 371 = (64.5+ m) x 3.5 rearrangement (1) m+64.5 = 371 / 3.5 evaluation of total mass (1) m+64.5 = 106 (kg) evaluation of woman's mass (1) m = 106-64.5 = 41.5 (kg)	full marks will be awarded for correct numerical answer without working	(4)

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) KE = ½ x 64.5 x 3.5^2 evaluation (1) 395 (J)	allow answers which round to 395 e.g. 395.0625 full marks will be awarded for correct numerical answer without working	(2)

Q10.

	Answer	Additional guidance	Mark
	correct data point(s) seen (1) (accel =) $\frac{\Delta v}{t}$ (1) evaluation (1) 2.9 (m/s^2)	allow MP1 and MP2 in either order any data point(s) on the line e.g. (1.4,4) allow 'gradient' allow e.g. $\frac{4}{1.4}$ for 2 marks allow values that round to 2.9 (m/s^2) (e.g. 2.857…) award full marks for the correct answer without working	**(3)** **AO3**